Does scale still help at 1.47B?

Train three smaller Qwen3 models on the exact default corpus and full schedule. Compare 255M, 498M, 985M, and the completed 1.47B control. Only depth varies: 4, 8, 16, and 24 layers.

Equal training exposure

232,090,905 documents; 248.584B raw tokens. 34,092,146 packed examples; 266,345 optimizer updates. Preserve exp277 optimizer, augmentation, shuffle, tokenizer and seeds.

Current state

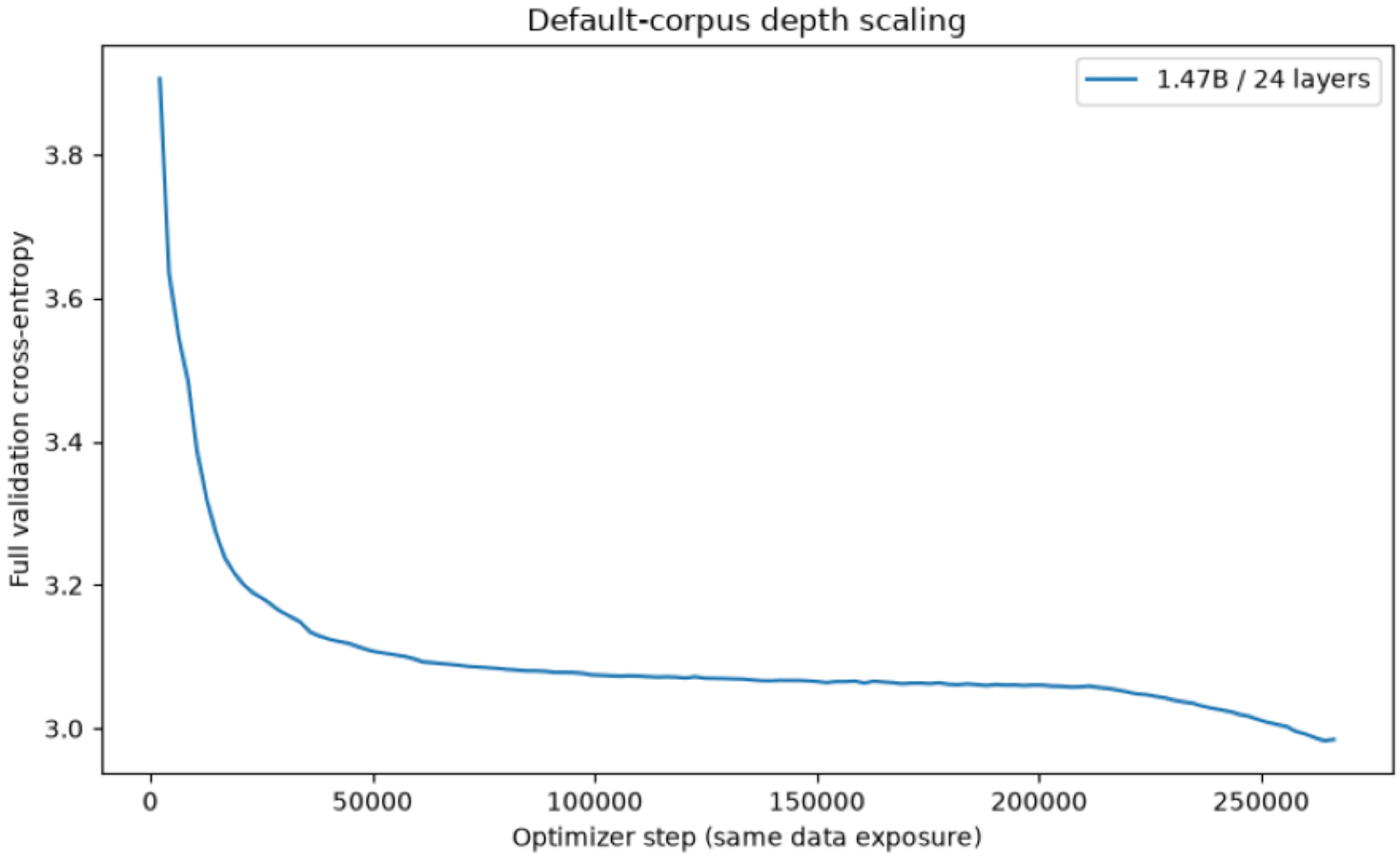
Experiment #362 registered. Resolved configuration and finite-data tests pass. TRC preflight found no complete regional MPNN caches. Proceeding with existing CoreWeave inputs at batch priority. All three smoke gates passed, including native/HF checkpoints and tokenizers. All three production runs submitted for all 266,345 updates, using 128 H100s. 255M and 985M have reached optimizer updates; 498M is starting. Full training and contact accuracy results are pending.

Interpretation

Use final validation loss and matched eval-val / eval-denovo contact accuracy. A fixed-recipe depth sweep can motivate larger runs, but cannot establish their performance. One seed per size; no held-out eval-test access.

validation_curves.png

Full validation curves at identical data exposure. Incomplete curves are not final results.




```
$ plot_training.py
```